

- 1) Login to EMR instance using Hadoop and then switch user to root using sudo -i.

Run the following command to instal MySQL connector jar file.

```
wget https://de-mysql-connector.s3.amazonaws.com/mysql-connector-java-8.0.25.tar.gz
```

```
tar -xvf mysql-connector-java-8.0.25.tar.gz
```

```
cd mysql-connector-java-8.0.25/
```

```
sudo cp mysql-connector-java-8.0.25.jar /usr/lib/sqoop/lib/
```

- 2) Ingest data from mySQL RDS to HBase table:

```
sqoop import --connect jdbc:mysql://hamariyusql.cgqj3fbcprpx.us-east-1.rds.amazonaws.com/taxi_records --username admin --table trip_log --hbase-table trip_log_hbase --column-family cf1 --hbase-create-table --hbase-row-key tpep_pickup_datetime,tpep_dropoff_datetime --hbase-bulkload --split-by payment_type
```

```
Map output bytes=41181317811
Map output materialized bytes=4892837443
Input split bytes=591
Combine input records=0
Combine output records=0
Reduce input groups=18842048
Reduce shuffle bytes=4892837443
Reduce input records=302089520
Reduce output records=301472768
Spilled Records=1104697432
Shuffled Maps =5
Failed Shuffles=0
Merged Map outputs=5
GC time elapsed (ms)=15178
CPU time spent (ms)=2876050
Physical memory (bytes) snapshot=4982210560
Virtual memory (bytes) snapshot=21423403008
Total committed heap usage (bytes)=4391960576

Shuffle Errors
BAD ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=23981007255
12:15:58 INFO mapreduce.ImportJobBase: Transferred 22.3341 GB in 2,314.3603 seconds (9.8818 MB/sec)
12:15:58 INFO mapreduce.ImportJobBase: Retrieved 302089520 records.
12:15:58 WARN mapreduce.LoadIncrementalHFiles: managed connection cannot be used for bulkload. Creating unmanaged connection.
12:15:59 INFO impl.MetricsConfig: loaded properties from hadoop-metrics2-hbase.properties
12:15:59 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
12:15:59 INFO impl.MetricsSystemImpl: HBase metrics system started
12:15:59 WARN mapreduce.LoadIncrementalHFiles: Trying to bulk load hfile hdfs://ip-172-31-36-93.ec2.internal:8020/user/hadoop/trip_log/cf1/6d40f40f7
ddal62bd054be99 with size: 11191064341 bytes can be problematic as it may lead to oversplitting.
12:15:59 WARN mapreduce.LoadIncrementalHFiles: Trying to bulk load hfile hdfs://ip-172-31-36-93.ec2.internal:8020/user/hadoop/trip_log/cf1/786db40a8
3077bdb3190c90 with size: 11191083018 bytes can be problematic as it may lead to oversplitting.
12:15:59 INFO Configuration.deprecation: hbase.offheapcache.minblocksize is deprecated. Instead, use hbase.blockcache.minblocksize
h-172-31-36-93 ~$
```

```
hbase(main):001:0> list
TABLE
trip_log_hbase
1 row(s) in 0.2730 seconds

=> ["trip_log_hbase"]
hbase(main):002:0> describe trip_log_hbase
NameError: undefined local variable or method 'trip_log_hbase' for #<Object:0x7f0d8eff>

hbase(main):003:0> describe 'trip_log_hbase'
Table trip_log_hbase is ENABLED
trip_log_hbase
COLUMN FAMILIES DESCRIPTION
(NAME => 'cf1', BLOOMFILTER => 'ROW', VERSIONS => '1', IN_MEMORY => 'false', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', TTL => 'FOREVER',
COMPRESSION => 'NONE', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0')
1 row(s) in 0.1030 seconds

hbase(main):004:0> count 'trip_log_hbase'
```